

# Cybersecurity Incident Report

## Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

It's a SYN FLOOD ATTACK.

The logs show that:

Someone is flooding our servers with a DOS attack causing issues for other users and eventually causing it to crash. This example features the TCP 3 way handshake process with users first establishing a connection and then requesting files from the server to load for example a webpage when you're online.

This event could be:

The most likely cause of this event is a SYN flood attack. The attacker from the IP address 203.0.113.0 is flooding the server (192.0.2.1) with SYN requests and sometimes getting the server to reply with SYN ACK. The weird thing is the attacker never receives the ACK back from the server. The handshake is never completed and the attacker continues to pile on requests that fill up the server's connection queue. It's repeated rapidly insinuating the an attempt to crash and freeze the network. The yellow highlighted logs represent signs of server stress from the attack. You can see the RST packets being sent to devices in an attempt to retry and reload the page but they aren't completed. The server is force closing connections because its resources are maxing out. When we see HTTP/1.1 504 Gateway Time-out it means the server is struggling to serve legit traffic.

## Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. First the device sends an SYN request to the server .
2. Server send [SYN, ACK] back and keeps a port open
3. Device send the Final ACK to complete the connection

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

The server essentially runs out of resources and ports to handle the amount of SYN packet requests coming at it. In response it either slows, freezes, or crashes.

Explain what the logs indicate and how that affects the server:

The logs show a SYN flood attack from IP 203.0.113.0, where the attacker keeps sending SYN requests without completing the handshake. This forces the server to hold open many half-connections, quickly using up resources and causing slowdowns or failures for real users. As a result, the server starts timing out requests and resetting connections, showing it's struggling to keep up.